

SKILL – 20

SIGTSTP and process suspension:

Code:

```
GNU nano 7.2 skill20.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <termios.h>
#include <errno.h>

volatile sig_atomic_t signal_received = 0;

void handle_usr1(int sig)
{
    signal_received = sig;
}

int main(void)
{
    pid_t child;
    int status;

    setbuf(stdout, NULL);

    printf("==== PART 1: SIGTSTP SUSPEND AND RESUME =====\n");

    [ Read 138 lines ]
G Help  W Write Out  W Where Is  C Cut  E Execute  C Location  U Undo  A Set Mark
X Exit  R Read File  A Replace  U Paste  J Justify  / Go To Line  M Redo  M Copy
```

Output:

ls -l output

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-20$ ls -l skill20.c skill20
-rwxr-xr-x 1 bhardwaj bhardwaj 16936 Oct 10 05:24 skill20
-rw-r--r-- 1 bhardwaj bhardwaj 3436 Oct 10 05:24 skill20.c
```

```
==== PART 1: SIGTSTP SUSPEND AND RESUME =====
Child process started. PID = 3261
Child is working; its state will be preserved.
Child working: step 1
Child working: step 2
Parent sends SIGTSTP to child PID 3261.
Child suspended by signal 20.
Parent sends SIGCONT to resume the child.
Child working: step 3
Child working: step 4
Child working: step 5
Child completed its work.
Child has finished; exit status collected.

==== PART 2: PROCESS GROUPS AND TERMINAL =====
Parent PID = 3260, child PID = 3264
Assigned child process group ID = 3264
Group child PID = 3264, PGID = 3264
Terminal foreground group temporarily transferred.
Terminal foreground group restored to parent group.
Parent sends SIGUSR1 to process group 3264.
Child received SIGUSR1 from its process group.
Process group signal test completed.
==== SKILL 20 COMPLETED =====
```

process information and terminal device output.

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-20$ ps -o pid,ppid,pgid,stat,cmd -C skill20
PID   PPID   PGID STAT CMD
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-20$ ps -o pid,ppid,pgid,ttty,stat,cmd -p $$
PID   PPID   PGID TT   STAT CMD
3187  3185   3187 pts/0  Ss   -bash
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-20$ tty
/dev/pts/0
```